

Faculty of Computers & Information Technology – Innovation University

Academic Year: 2025/2026

Database Project Proposal Form

Project Title: Sustainable Waste Management and Recycling System

TA Name: -

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Statement of Problem:

Many communities lack an organized system for managing recyclable waste and reusable materials. As a result, large amounts of waste are improperly disposed of, leading to environmental pollution and loss of valuable resources. Additionally, individuals and organizations often find it difficult to connect with recycling centers or track their recycling activities. This creates inefficiencies and reduces participation in sustainable practices. This project aims to address these issues by developing a database system that organizes waste collection, recycling processes, and user participation.

Brief Description of the Project:

The system is a database-driven application that allows users to register recyclable materials, request pickups, and track their recycling history. It stores data related to users, waste types, collection requests, and recycling centers. The system helps improve

coordination between users and recycling facilities while promoting environmentally friendly practices.

Objectives:

- To organize and store data related to recyclable materials and users
- To encourage recycling and reuse practices
- To improve waste collection efficiency
- To connect users with recycling centers
- To generate reports on recycling activities and environmental impact

Technical Approach:

- Design ER Diagram (ERD)
- Create normalized database tables
- Use SQL Server as the DBMS
- Implement SQL queries (CRUD operations)
- Apply constraints and validation
- Develop a simple user interface (optional)

Expected Deliverables:

- Database schema (tables and relationships)
- ER Diagram (ERD)
- SQL scripts
- Sample data
- Documentation/report
- Presentation